

## TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 1

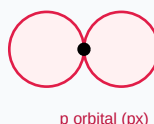
# CHAPTER 4: TRANSITION AND INNER TRANSITION ELEMENTS

## PREMIUM ACADEMIC STUDY SUITE &amp; EXAM GUIDE

|               |                                             |
|---------------|---------------------------------------------|
| Board:        | Tamil Nadu State Board (TNSB)               |
| Syllabus:     | Samacheer Kalvi Textbook Curriculum aligned |
| Standard:     | Class 12 English Medium                     |
| Subject:      | Chemistry Volume 1                          |
| Chapter Name: | Transition and Inner Transition Elements    |

**Learning Objectives:**

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

**Atomic Orbitals Wave Functions (s & p)**

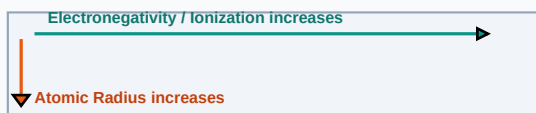
## Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

| Term                   | Official Definition & Context                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Transition Element     | An element whose atom has a partially filled d subshell in ground state or common oxidation states.                                           |
| Lanthanide Contraction | The steady decrease in atomic and ionic radii of lanthanide elements with increasing atomic number, caused by poor shielding of 4f electrons. |
| Interstitial Compound  | Compounds formed when small atoms like H, C, or N are trapped in the interstitial spaces of transition metal lattices.                        |

**Syllabus Exam Tip:** Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

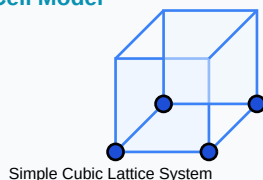
### Periodic Properties Trends Direction Map



## Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

### Crystal Solid Lattice Unit Cell Model



### Essential Formulas, Laws & Identities:

- Magnetic Moment:  $\mu = \sqrt{n(n+2)}$  Bohr Magnetons ( $n$  = unpaired electrons)
- Lanthanide shielding sequence:  $s > p > d > f$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

## Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

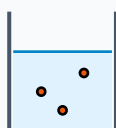
**Example 1: Calculate spin-only magnetic moment of  $\text{Fe}^{2+}$  ( $Z=26$ ).**

**Step-by-step Solution:**

Fe:  $[\text{Ar}] 3d^6 4s^2$ .  $\text{Fe}^{2+}$ :  $[\text{Ar}] 3d^6$ . Orbitals have 4 unpaired electrons ( $n=4$ ).

$\mu = \sqrt{4 \times 6} = \sqrt{24} = 4.90$  Bohr Magnetons.

### Titration Beaker & Solute Precipitation



Reacting Mixture

## HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

**HOTS Challenge 1: State Lanthanide Contraction. Explain its causes and two key consequences on transition elements properties.**

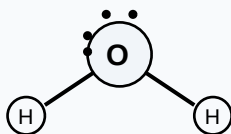
**Analytical Solution Strategy:**

Steady radius decrease from La to Lu.

Cause: Poor shielding of 4f electrons fails to counter growing nuclear charge.

Consequences: (1) 4d and 5d transition series (e.g. Zr and Hf) show almost identical atomic sizes, (2) Basic strength of hydroxides decreases from  $\text{La}(\text{OH})_3$  to  $\text{Lu}(\text{OH})_3$ .

**Lewis Dot Structure (H<sub>2</sub>O Bent Molecular Geometry)**



## Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

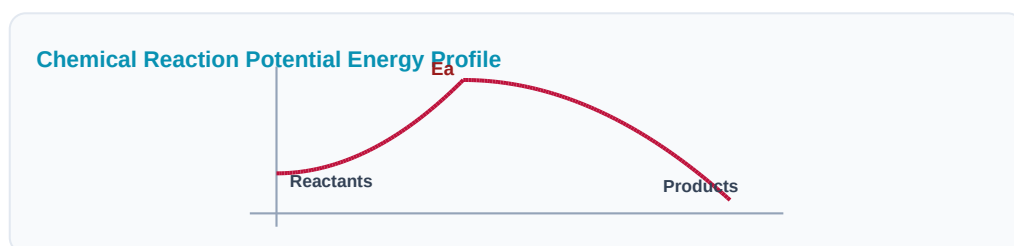
**Q1 (1-Mark):** Why do transition metals exhibit variable oxidation states?

**Q2 (2-Mark):** Explain why transition metal complexes are colorful.

**Q3 (2-Mark):** Describe the preparation of potassium dichromate ( $K_2Cr_2O_7$ ) from chromite ore.

**Q4 (5-Mark):** List three differences between lanthanides and actinides.

**Q5 (5-Mark):** Why is zinc not considered a true transition metal?



## Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

### Application Case: Aerospace Alloys

Titanium transition metal lattices are alloyed with aluminium to build jet turbine blades.

### Application Case: Industrial Catalysis

Finely divided nickel and platinum speed up food hydrogenation and auto emission filters.

**Cross-Curricular Link:** Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

#### Aromatic Benzene Ring (C<sub>6</sub>H<sub>6</sub>) Resonance



## Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

### Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla,  $\text{cm}^3$ , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

### Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

**CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!**

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.  
Good luck!

#### Electrochemical Galvanic Cell System

